



The Oxford Handbook of Cognitive Psychology

Daniel Reisberg (ed.)

<https://doi.org/10.1093/oxfordhb/9780195376746.001.0001>

Published: 2013

Online ISBN: 9780199978304

Print ISBN: 9780195376746

CHAPTER

9 Disorders of Attention

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<https://doi.org/10.1093/oxfordhb/9780195376746.013.0009> Pages 131–146

Published: 03 June 2013

Abstract

Attention covers a wide range of different concepts, variously referred to as spatial attention, divided attention, sustained attention, focused attention, visual attention, covert mental attention, inhibition of attention, and many others. There are likewise diverse disorders of attention resulting from different forms of brain damage. This chapter discusses forms of unilateral spatial neglect following unilateral brain damage as disorders of spatial attention and of visuospatial working memory, specific disorders of divided and sustained attention associated with Alzheimer's disease, and failures of inhibition and focused attention associated with focal brain damage in the prefrontal cortex. Behavioral and neuroanatomical evidence is considered along with accounts of the cognitive impairments observed and implications of this evidence for theories of healthy cognition.

Keywords: visuospatial working memory, unilateral spatial neglect, representational neglect, Alzheimer's disease, frontal lobe damage, brain damage

Subject: Cognitive Psychology, Psychology

Series: Oxford Library of Psychology

Attention conjures up a range of concepts in the lay public and in scientists alike. In everyday use, attention generally refers to the idea that humans have some kind of limited cognitive resource, and attention involves concentrating at least some of that resource on a specific task or a narrowly defined topic, or on a single source of information. However, this concept of attention can apply to quite complex tasks such as “paying attention while driving,” or it can be rather narrow such as attending to the ticking of a clock or to a particularly attractive face to the exclusion of anything else around you. Attention can also apply to mental objects and events such as how to answer a difficult question in a job interview, the memory of a favorite landscape, or remembering to carry out each of a set of errands in the shopping mall.

Theories of attention in healthy adults are covered elsewhere in this volume (see Part 2). Disorders of cognition are generally better understood in the context of theories of cognition, and attention is no exception. Moreover, there are numerous instances in which testing theories of cognition when studying brain-damaged individuals has led to refinement of those theories. So studying the damaged brain can help

us understand aspects of the healthy brain as well as understand the nature of the cognitive impairments associated with the damage. In this chapter, I will cover disorders of spatial attention, commonly associated with relatively focal right hemisphere lesions. I will also cover disorders of divided attention linked with degenerative brain diseases such as the dementias, and that are associated with more diffuse damage. Finally, there will be some discussion of disorders of sustained attention and of failures of inhibition that are often associated with damage to areas in the prefrontal cortex.

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Disorders of Spatial Attention

Unilateral Spatial Neglect

Unilateral spatial neglect refers to an apparent lack of awareness of the side of space opposite to the brain hemisphere that is damaged. Moreover, patients seem to be unaware that they have this disorder, even when it is pointed out to them. It is not uncommon following a stroke or a brain injury. Two months post onset, 48% of right brain-damaged (left neglect) and 15% of left brain-damaged (right neglect) patients show neglect (Halligan, Marshall, & Wade, 1989; see also Vallar & Perani, 1986). Neglect of the left side is not only more frequent than right neglect (e.g., Kinsbourne, 1987), it is also more severe and tends to last longer (e.g., Stone et al., 1991). However, there is an ongoing debate as to its interpretation, largely because there are several different types of neglect but also because there are no clear links between the precise site of the lesion and the form of neglect that is observed. Indeed, neglect might arise from disruption of networks within each hemisphere of the brain, rather than from damage to any one particular site. In the discussion that follows I will describe the different forms of the disorder and the contrasting views as to their interpretation.

Perceptual Neglect

The most widely studied forms of neglect involve a tendency to ignore one-half of extrapersonal space, most commonly on the left, following a lesion in the right parietal lobe (e.g., Paterson & Zangwill, 1944; for reviews, see Robertson & Halligan, 1999). These neglect types are linked with problems of perceiving the external world and so are referred to as perceptual neglect (Albert, 1973; Poppelreuter, 1917/1991). Thus, when asked to cross out items on a sheet in front of them, to copy a picture, or to indicate the center point of a line, patients with perceptual neglect will cross out items mainly on the right, reproduce only the right half of the picture, or will bisect a line well to the right of the true center (e.g., Schenkenberg, Bradford, & Ajax, 1980). When asked to describe their immediate environment, they will describe only the right half of the scene in front of them, and, in some cases, will eat from only the right half of their plate or fail to comb their hair or shave on their left side. For some patients, they show what is called “neglect dyslexia” in which they fail to read the left half of individual words (e.g., Miceli & Capasso, 2001). Others show the phenomenon of “allochiria” in which an item that actually appears on the left is reported by the patients as having been presented on the right (e.g., Lepore, Conson, Ferrigno, Grossi, & Trojano, 2004). These deficits do not arise from failure of the visual sensory system such as poor vision or hearing loss, nor from a general intellectual impairment. The problem appears to arise from a failure of visual attention to areas of space.

There are multiple forms of perceptual neglect, for example, some patients show “personal” neglect, ignoring one side of their body, while others show “extrapersonal” neglect showing that they are not aware of the left side of space (e.g., Beschin & Robertson, 1997). Even within extrapersonal neglect, there are cases of neglect of the immediate environment (near space) but not of more distant aspects of the current scene (far space) (e.g., Calvanio, Petrone, & Levine, 1987; Halligan & Marshall, 1991). These diverse patterns among patients considered under the general umbrella of unilateral spatial neglect led Halligan and

Marshall (1992) to question whether this classification of the disorder was “a meaningless entity.” As we will see, this challenge contains an important kernel of truth, because there is diversity among the types of neglect, as well as a diversity of accounts for each type. Whether this renders the broader classification “meaningless,” however, remains moot.

Although there are several different forms of neglect, one common interpretation is that they reflect an inability to attend to the neglected side of space or an inability to move attention away from the right (e.g., De Renzi, Gentilini, Faglioni, & Barbieri, 1989; Posner, Walker, Friedrich, & Rafal, 1984). In terms of underlying brain organization, Kinsbourne (1987) has suggested the hypothesis that the preponderance of left neglect following a right hemisphere lesion arises because the right hemisphere represents both the left and right sides of space, but the left hemisphere represents only the right side of space. Therefore, when the left hemisphere is damaged, the intact right hemisphere can continue to represent both right and left hemispace. When the right hemisphere is damaged, the left hemisphere can only deal with the right side of space, leaving the left side of space without representation in the brain. However, this hypothesis does not deal adequately with the observation that some patients do show right neglect following left hemisphere damage and an intact right hemisphere. A more recent view from Bartolomeo, Thiebaut de Schotten, and Doricchi (2007) has argued that left neglect is not so much linked with damage to specific brain areas within the right hemisphere, but more that damage in those areas disrupts a network of brain white matter that is used for communication between the prefrontal cortex and the parietal areas in the right hemisphere. It is often the latter that are damaged in neglect.

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Some theoretical developments have focused on findings that aspects of neglect can be overcome on a temporary basis by providing external cues. So, for example, a number of studies have shown that line bisection in neglect patients tends to be much more accurate for very long lines than it is for short lines (e.g., Bisiach, Pizzamiglio, Nico, & Antonucci, 1994). Rossetti et al. (1998) showed that when neglect patients were asked to wear rightward displacing prismatic goggles, then they showed a dramatic reduction in a range of neglect symptoms. Although the benefit is clear, the explanation is much less so. One view is that perceptual neglect is a disorder of spatial attention, but it is uncertain whether this is an overemphasis on attention to the nonneglected side or an inability to direct attention to the neglected side. In either case, prism adaptation has been thought to help the patient initiate control of spatial attention to the neglected side (for a discussion see Nijboera, McIntosh, Nys, Dijkerman, & Milner, 2008). An alternative account is that the prisms help the patient to orient more effectively toward external cues in the environment. For example, a series of studies by Robertson and colleagues (e.g., Robertson, Mattingley, Rorden, & Driver, 1998) has shown that presenting an “alerting” signal such as a tone can result in neglect patients noticing events on the neglected side that follow the tone. They have argued that the alerting stimulus activates remaining healthy cells in the damaged right hemisphere that support spatial attention, and that these cells do not operate effectively without the external cue.

As a contrast to these findings that an external cue might alleviate symptoms of neglect, in “extinction,” the presence of a second stimulus is the source of the problem. In these cases, the patient can readily detect a single stimulus in his or her right or left visual field. However, if two stimuli are presented, one in each hemifield, then one of the stimuli appears to be ignored or is “extinguished” by the presence of the second stimulus (e.g., Humphreys, Riddoch, Nys, & Heinke, 2002; Valler, Rusconi, Bignamini, Geminiani, & Perani, 1994). Extinction can appear with other senses, such as hearing, touch and smell (e.g., Bender, 1952) and analogous methods are used to diagnose extinction in those other modalities. It is related to neglect in that the disorder appears to be a failure to attend to a stimulus in a particular area of space. However, extinction tends to appear only with brief presentations of the two stimuli, such as the clinician holding up a finger in the left or right visual field or in both fields of the patient. Unlike neglect, extinction shows no preferential association with the right hemisphere, and it may follow damage to either side of the brain, affecting the

side of space opposite to the damage. Its causes are not entirely clear, but like neglect, extinction appears to reflect the outcome of a spatially biased competition for attention.

Another line of study has explored whether some of the problems in neglect arise from an inability to keep track of scanning of visual scenes. Specifically, patients cannot remember which areas of a scene they have scanned and which they have not. Recordings of eye movements have shown that neglect patients tend to refixate the same items in an array that they have fixated before. When they are given an array of objects on a computer screen and asked to cancel as many items as they can within the array, they tend to cancel the same items on the right side repeatedly (Husain, Mannan, Hodgson, Wojciulik, & Driver, 2001). This has been interpreted as a failure of visuospatial working memory to retain and update a representation of what has and what has not been searched in an array. Husain et al. (2001) suggest further that impaired spatial working memory is the basis for pathological visual search observed in several patients with neglect, and they propose a possible link between visual spatial attention and visuospatial working memory (Logie, 1995). Some supportive evidence for this comes from studies of healthy adults showing that when people are asked to follow a moving target on screen, this disrupts their ability to remember spatial arrays (Pearson & Sahraie, 2003; Postle, Idzikowski, Della Sala, Logie, & Baddeley, 2006). This is then consistent with there being a link between visual attention and spatial working memory in the healthy brain, and with the idea that this link is damaged in at least some forms of neglect. However, it is not clear whether the Husain et al. (2001) finding is linked primarily with perceptual neglect, or if their patients had both perceptual and representational neglect (see later discussion), with the latter being at the core of the problem with visuospatial working memory.

Representational Neglect

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The debate continues with regard to explanations for perceptual neglect, but the issue is not made any simpler by observations of what has been referred to as representational neglect. The best known report is of two neglect patients living in Milan, Italy. Bisiach and Luzzatti (1978) asked these patients to try to imagine that they were standing in the main cathedral square in Milan (Piazza del Duomo), an area that they knew well. When asked to imagine that they were at one end of the square, facing the cathedral, they mentioned a range of buildings that are on the right side of the square, but they mentioned very few buildings from their imagined left. However, this was not because they had forgotten these details. They were next asked to try to imagine themselves at the other end of the square, with their back to the cathedral. This time, they reported buildings now on their imagined right that they had previously failed to mention, and they failed to report buildings now on their imagined left that they had mentioned before. So the problem was with the left side of their mental image or spatial mental representation of the square, regardless of their imagined viewpoint. The patients showed the same lateralized impairments of mental images when asked to describe a room in their home, so the effect seems to generalize to any familiar setting.

In the Bisiach and Luzzatti (1978) cases, the patients showed evidence of perceptual neglect for the left side of physical space as well as neglect of the left side of imagined space. However, there are multiple reports of patients who show only perceptual forms of neglect but have no difficulty describing familiar scenes from memory (e.g., Coslett, 1997). There are also reports of patients who show

Figure 9.1

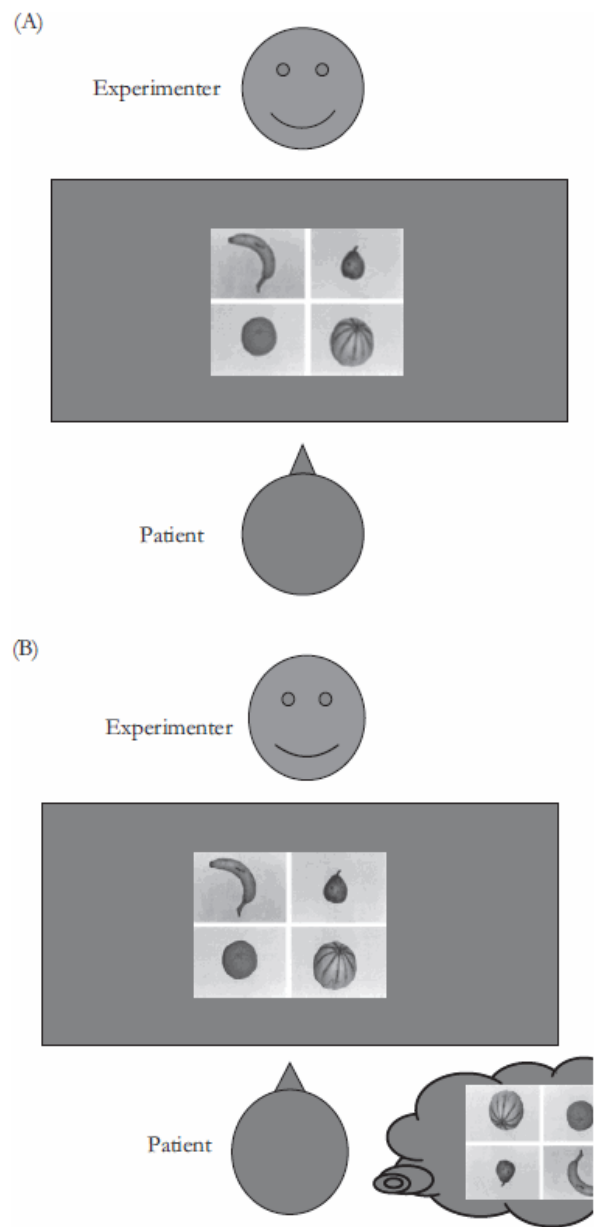


Example drawing of a familiar country scene by patient NL (Beschin et al., 1997), who suffered from pure representational neglect.

a pure form of representational neglect (Beschin, Cocchini, Della Sala, & Logie, 1997; Guariglia, Padovani, Pantano, & Pizzamiglio, 1993). These patients show no symptoms of perceptual neglect in that they readily notice objects that are in view regardless of whether they are on the left or on the right, and they show no impairments in item cancellation tasks or line bisection. However, when asked to describe a familiar scene from memory, they fail to report details from the left side of their imagined perspective. Figure 9.1 shows an attempt by patient NL (Beschin et al., 1997) to draw a familiar country scene from memory. What is clear is that he drew considerable detail on the right of the picture and drew almost no details on the left. When questioned about this, he could see in the drawing that there were few details on the left, but he responded that there were very few details in that part of the real scene. His friends and family confirmed that there were indeed as many details on the left as on the right of this area of countryside, which he had visited many times. So he could see the lack of detail in his own drawing, showing no perceptual neglect, but he could not generate the required details from his memory for the scene. Cases like this point to the suggestion that perceptual neglect and representational neglect do not reflect variations of the same disorder, and the fact that both kinds of patient have been observed suggests it is not simply that, for example, more severely affected patients have both forms but less severely affected patients only have the most common, perceptual form.

- p. 135 Pure representational neglect patients are much less commonly described. This could be in part because tests for representational neglect are rarely used in clinical assessment of stroke patients. However, such patients do appear to be less common even when neuropsychologists are specifically looking for them. In a group study of 17 right hemisphere–damaged patients, Denis, Beschin, Logie, and Della Sala (2002) found one patient with pure representational neglect, eight patients with both representational and perceptual neglect, and two patients with only perceptual neglect. Six of the patients had no neglect, showing that neglect is not inevitable following right hemisphere damage.

Figure 9.2



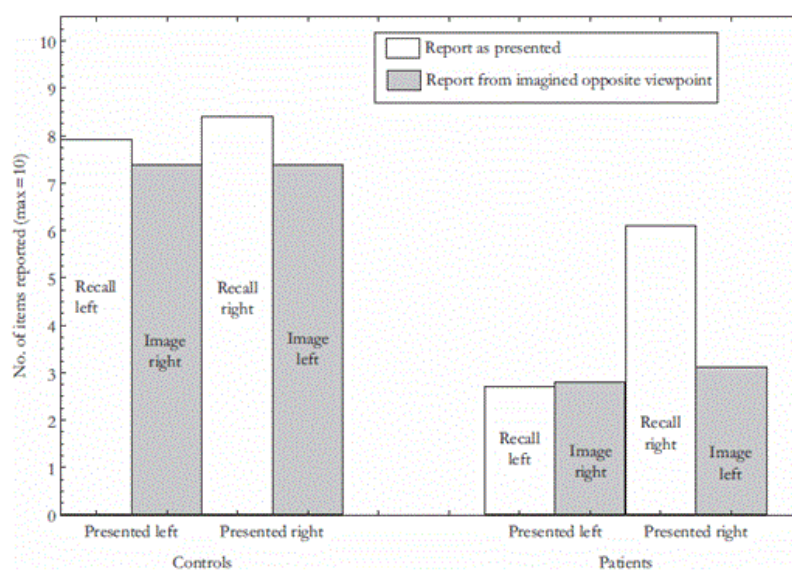
Cartoon illustration of the layout for testing perceptual and representational neglect with patients reporting the layout as viewed (A) or from the perspective of the experimenter (B).

Moreover, there was no relationship between the precise site or the extent of brain damage and the type of neglect, consistent with the idea that different forms of neglect might arise from disruption to communication between brain areas rather than damage to any one specific area.

The Denis et al. (2002) study demonstrated another phenomenon that seriously undermined any suggestion that representational neglect somehow arises from some aspect of neglect in perception or visual attention. Denis and colleagues tested their patients on novel arrays of objects. Four objects were presented in a simple square layout as illustrated in Figure 9.2A. In the perception condition, participants simply had to report what they saw while the objects were still in view, naming each object and its location in the array. In a memory condition, the array was removed after viewing and the participant was asked to recall from memory each of the objects and their location. In a third condition, the array was not shown to the participant, but instead they heard the layout of objects described by the experimenter. They then had to

recall from memory each object and its location. Results showed that in the perception condition, the patients with both perceptual and representational neglect tended to report either zero or only one item from the left of this very simple array, but they could report both objects on the right. In the memory condition, the pattern of results was the same, except that overall performance was a bit lower. Most intriguing, the patients also reported fewer items on the left of the array that they did not see but only heard described. The patients had completely intact verbal immediate memory, so it seemed as though they were trying to remember the verbally described arrays using a damaged visuospatial working memory system, despite the fact that they could have remembered the items quite successfully by using their intact verbal memory. This points very clearly to the suggestion that representational neglect arises for different reasons than perceptual neglect. Moreover, because

Figure 9.3



Results from Della Sala et al. (2004) for a group of controls and two representational neglect patients reporting the layout of four objects as viewed or after rotation through 180 degrees.

representational and perceptual neglect are separate impairments (Coslett, 1997), this suggests that the visual attention system that supports perception of the world in the healthy brain is not the same as the system that supports our visual mental representations (Logie, 2003, 2011; Logie & van der Meulen, 2009). If they were the same, then damage to one should automatically result in damage to the other, and clearly this is not the case.

Can representational neglect be interpreted as an impairment in some form of covert or mental attention to parts of our mental representation? Logie, Beschin, Della Sala, and Denis (2005; Della Sala, Logie, Beschin, & Denis, 2004) addressed this question with two representational neglect patients who were given the same range of experimental conditions used by Denis et al. (2002). But in addition, there were conditions in which each patient was asked to report the location of each of the four objects as they would appear to the experimenter on the other side of the table (see Fig. 9.2B). Results from this experiment are shown in Figure 9.3. First, note that the neglect is clear in both patients when asked to recall the objects in their original locations (plain bars). However, when they were asked to imagine from the opposite perspective (i.e., mentally rotate their visual representation—shaded bars in Figure 9.3), there is no further loss of items presented on the left but now recalled from the imagined right. However, for items presented on the right and recalled from the imagined left (the neglect side), performance drops to the level it would have been if the items had been presented on the left in the first place. Logie et al. (2005) interpreted these results to

suggest that the patients could pay “mental attention” to material on both sides of the representation, and they could also control mental attention in order to carry out the mental rotation required. However, they performed poorly on the basis of whether the item that they were recalling was on the left of their mental representation. This pointed to an impairment in the representation system in visuospatial working memory, not to the control of mental attention. A possible analogy is that the left half of the “mental screen” was damaged, and that an intact covert attention system was dealing with a mental image that was impoverished on one side. Consistent with this interpretation, Baddeley and Lieberman (1980) speculated that representational neglect might reflect a deficit in the visuospatial system within working memory, a suggestion that was also raised by Bisiach (1993).

A further case illustrates that perceptual and representational neglect are really rather distinct. Bescin, Basso, and Della Sala (2000) reported the case of a very unfortunate patient who suffered strokes in both hemispheres. This patient showed both perceptual and representational neglect, but what was intriguing about this case was that he showed the two forms of neglect in different hemifields. Figure 9.4 shows attempts by the patient to draw a picture of a house shown at the top of the figure. The middle of the figure shows the patient’s attempt to copy the picture of the house when it was still in view, and it is clear that he was copying only the left half of the picture (a right perceptual neglect). The bottom drawing is the attempt by the patient to reproduce the house from memory, and it is clear that he was recalling only the details from the right half of the house (a left representational neglect). The fact that his copying was intact shows that the absence of the left half of the house when drawing from memory was not due to a failure to attend to the left half of the drawing when it was in view. Finding the two forms of neglect independently for different hemifields in the same patient offers rather strong evidence that they reflect very different disorders. This reinforces the view that visual attention and perception rely on different neuronal pathways in the healthy brain than

Figure 9.4



Drawing from copying (*middle*) or from memory (*lower*) of a picture of a house by a patient suffering from right perceptual neglect and from left representational neglect (Reproduced from Beschin et al., 2000, with permission.)

visuospatial working memory and covert attention to mental processes.

Implicit Processing in Neglect

p. 138 A final observation regarding neglect comes from a range of studies that demonstrate some form of implicit processing of material in the neglected field, at least in some patients. The most famous of these cases was reported by Marshall and Halligan (1988), in which a patient was shown a picture of two houses, one above the other. One of the houses looked perfectly symmetrical and normal, while the other house was depicted with flames issuing from a window on the left of the picture. ↪ The patient was asked to indicate if she detected whether the two houses were the same, or were different in any way. She consistently reported that the houses were identical. However, when the patient was asked to choose in which house she would prefer to live, she tended to choose the house without the flames while claiming that this was a complete guess. Similar conclusions were drawn from studies using chimeric pictures that show the right half of one object/animal joined to the left half of another object/animal (e.g., Buxbaum & Coslett, 1994; Cantagallo and Della Sala, 1998; Young, Hellawell, & Welch, 1992). Some neglect patients demonstrated that they were implicitly processing the left half of the stimulus while maintaining that the two pictures were identical despite their chimeric nature. For example, Cantagallo and Della Sala's (1998) patient was presented with the drawing of a trombone with a rifle butt coming out from its left side. Spontaneously, he reported that he was seeing a "trombone which *fired* notes."

Recently, Della Sala, van der Meulen, Bestelmeyer, and Logie (2010) showed pictures to neglect patients in which the neglected half depicted a scene linked to either a well-known proverb such as "Don't put all your eggs in one basket" or to an unfamiliar proverb translated into English from other languages such as "Don't buy a cat in a bag" (see Figs. 9.5A and 9.5B). After looking at the picture, it was clear that each patient was not aware of the content of the left side. The experimenter then read out a series of proverbs, only one of which was semantically linked to the neglected half of the picture (printed in *italic* in Figs. 9.5A and 9.5B). Although the patients maintained that they were guessing, they successfully performed above chance in selecting the correct proverb, but only when proverbs were familiar. When proverbs were unfamiliar (such as the translated Dutch proverbs in Fig. 9.5B), performance was at chance. In that same study, a group of healthy participants was asked to carry out the same task, except that the target area of each picture was presented very rapidly, and below the perceptual threshold for each participant to be able to detect what had been presented. They showed the same pattern as the patients in that they could successfully select the correct proverb from a group of familiar proverbs, but they could not do so when the proverbs were unfamiliar.

p. 139 These studies of implicit processing suggest that although neglect patients are not able to attend to the left half of their visual field, information from the neglected field is being processed at a level below conscious awareness. The flames of the burning ↪ house in the Marshall and Halligan (1988) study appear to activate stored knowledge in the patient about the undesirability of a house on fire. The selection of the proverbs in the Della Sala et al. (2010) study suggests that the activation from the neglected field is sufficient to support quite complex semantic processing. That is, even when attention is impaired, this does not mean that the "unattended" material is not processed by the brain, possibly even at the semantic level. Other studies of neglect patients have shown that they can avoid obstacles when reaching or moving, even when those obstacles are in the neglected field and the patient would fail to report those objects if asked to describe what he or she can see (e.g., McIntosh et al., 2004).

Figure 9.5A



Still waters run deep
Blood is thicker than water
Don't put all your eggs in one basket
Walking on egg shells

Example of a picture used by Della Sala et al. (2010), with a target feature on the left of the picture to be matched with a *familiar* proverb from among those shown. The correct proverb is shown here in bold italics, but it was not highlighted in any way in the actual experiment.

Figure 9.5B



Example of a picture used by Della Sala et al. (2010), with a target feature on the left of the picture to be matched with a *familiar* proverb from among those shown. The correct proverb is shown here in bold italics, but it was not highlighted in any way in the actual experiment.

Finally, the finding that there are patients with perceptual neglect but not representational neglect suggests that a disorder of visual attention will not necessarily imply that there is a disorder of covert attention to mental events within working memory. Conversely, the finding that there are patients with representational neglect who do not have perceptual neglect suggests that the representational neglect is not the direct result of a disorder of visual attention, and it does not appear to be a disorder of covert attention. Perceptual and representational neglect often occur together in stroke patients, but they do appear to reflect a combination of two different disorders that have rather different characteristics. Although lesion sites in neglect patients tend to be in the right fronto-parietal areas of the brain, as I have noted earlier in this chapter, whether there are specific areas of the brain linked with one or other form of the disorder is not clear. For example, in the Denis et al. (2004) study, there was no relationship between the locus or extent of the lesion in each patient and whether they showed perceptual or representational neglect or no neglect at all. A plausible hypothesis then is that it is the way in which the lesion affects connectivity (Bartolomeo et al., 2007) between the frontal and parietal lobes that leads to the form of cognitive impairment rather than the specific area of brain that is damaged by the lesion.

Disorders of Divided Attention

While the disorder in neglect, at least in perceptual neglect, appears to be linked with impairments of attention to locations in space, the attentional disorder in people with Alzheimer-type dementia appears to be a problem of dividing attention. Alzheimer-type dementia is linked with a brain disease that gradually spreads through the brain. It most commonly (but not always) starts in the temporal lobes, and as the disease progresses, it affects other brain areas, notably the prefrontal cortex. The precise link between the areas of brain affected by the disease and the cognitive impairments from which these patients suffer is not entirely clear. Nevertheless, the cognitive impairments in such patients have been studied quite extensively, and episodic memory failure is a common early symptom of the disease, linked with the temporal lobe damage. However, impairments of episodic memory are also found in healthy older people and in people with other disorders such as chronic depression. So a major criterion for diagnosing Alzheimer's disease is that the person is shown to have impairments in episodic memory and in at least one other cognitive function, and that these cognitive functions get poorer over time, indicating that the disease is progressing. One particular "other" cognitive function that appears to be impaired in people with this disease is control of attention (e.g., Baddeley, Logie, Bressi, Della Sala, & Spinnler, 1986; Perry & Hodges, 1999), and specifically, the ability to divide attention in order to carry out two tasks at the same time, or dual tasking.

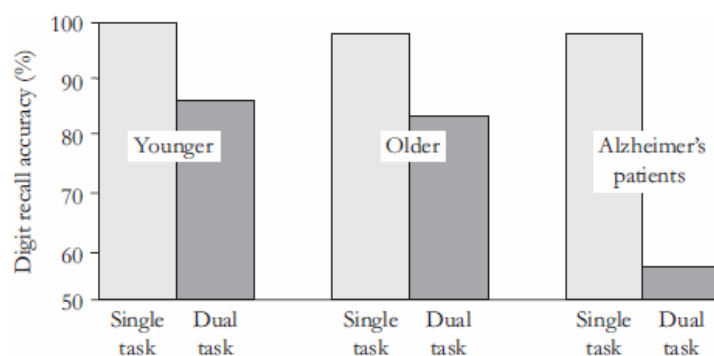
Divided Attention in Dementia

Dual tasking in healthy adults tends to be studied broadly in three ways. One of these investigates when two tasks compete for a common processing stage, such as dealing with two simultaneous inputs or having two competing responses (for reviews, see Levy, Pashler, & Boer, 2006; Pashler & Johnston, 1998). A second approach focuses on the ability to switch back and forth between two tasks that have to be performed in a limited time period (e.g., Allport & Wylie, 2000; Monsell, 2003). However, the dual-task problem that is specific to Alzheimer's disease arises from experimental paradigms that explicitly avoid perceptual or response bottlenecks. Instead, the two tasks are chosen so that they can be performed largely in parallel by healthy adults (Cochini, Logie, Della Sala, & MacPherson, 2002; Logie, Cocchini, Della Sala, & Baddeley, 2004).

The first study to address divided attention in Alzheimer's disease was reported by Baddeley et al. (1986). A group of patients who had already been diagnosed as having the disease using formal clinical criteria were compared with a group of older healthy people matched on age with the patient group, and with a third group of younger healthy people. One difficulty with comparing different groups on their ability to divide attention between two tasks is that the groups might differ in their ability to do each task on its own. So, for example, Alzheimer's patients have an episodic memory deficit, and so their digit span will be lower than it is for healthy people of the same age. Older healthy people, in turn, might have a lower digit span than younger healthy people. So, if the experiment compares the groups on how well they can recall digits while they are doing something else at the same time, then the patient group will perform more poorly simply because they had trouble when even doing the single task. The dual-task impairment could then be explained as an episodic memory impairment. To get around this problem, Baddeley et al. (1986) adjusted the demand of each of their two tasks according to the ability of each participant. Each individual in each group was first assessed for his or her immediate serial order verbal memory with aural presentation of sequences of digits that the participant was asked to recall orally in the same order. The length of the sequence on each trial was gradually increased until the individual could no longer recall the sequence correctly, and this was taken as the individual digit span, or a measure of the individual maximum capacity for retaining a verbal ordered sequence. The second task involved asking participants to use a light-sensitive stylus to follow a moving single target around a computer screen. The computer recorded when the participant was on or off the target, and the speed of the target was gradually increased until people

could no longer reliably keep the stylus on the target. This maximum speed for reliable tracking was taken as tracking span, or the maximum capacity for perceptual-motor tracking for each individual. This technique of titrating or adjusting demand for each person meant that everyone in all three groups was then performing at close to 100% of his or her own maximum ability on each task. That is, the initial, single task performance levels were the same across the three groups. So, when participants were asked both to recall digit sequences and to track the moving target at the same time, any differences between the groups in dual-task performance have to be specific to dual task, and not due to differences in initial baseline levels of performance. The results of this study for digit recall are shown in Figure 9.6.

Figure 9.6



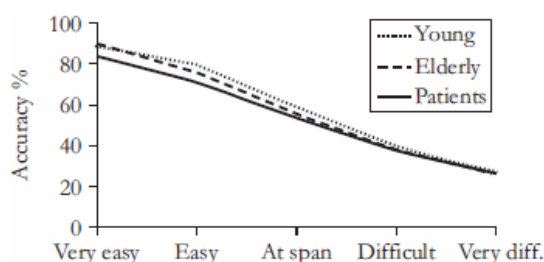
Digit recall alone and with concurrent tracking for younger and older people and for people with mild Alzheimer's disease. (Based on data from Baddeley et al., 1986.)

It is clear from Figure 9.6 that all three groups were indeed performing at their own maximum level when asked to perform digit recall on its own. However, when asked to listen to and recall digit sequences and at the same time to follow a moving target at their own individual maximum speed, then the healthy young and older adults showed about the same drop in performance of around 15%–20%, but the Alzheimer group show a very dramatic drop of around 45% compared with their own single task performance. A very similar result was found for the tracking performance in single and dual task. As noted earlier, this difference for the patient group cannot be attributed to initial differences in single-task performance between the groups, and so it appeared that the patients had a specific problem in carrying out two tasks at the same time. In a follow-up study, Baddeley, Bressi, Della Sala, Logie, and Spinnler (1991) tested the patients on three occasions, 6 months apart. The specific difficulty with doing two tasks at once became progressively larger as the disease progressed, again linking this dual-task impairment with the disease.

One possible criticism of the dual-task studies described earlier is that doing two tasks at the same time could be seen as simply more difficult or more cognitively demanding than doing one task at a time. This criticism arises from suggestions that cognition slows down in older people (e.g., Salthouse, 1993), and so it might be slowed down even further in people with progressive brain disease. A more cognitively demanding task will be more affected in people with progressive cognitive slowing as might be the case with Alzheimer patients. To address this criticism, Logie et al. (2004) used a variation of the procedure used in the earlier Baddeley et al. (1986) studies. Again, the individual spans for each participant were assessed for digit recall and for tracking. However, Logie et al. (2004) next asked participants in each group to perform the digit recall on its own while the length of the sequence that they had to recall was reduced or increased. They then asked participants to perform the tracking as a single task as the speed of the target was increased or decreased. Results for the tracking task are shown in Figure 9.7, and very similar for digit recall. When the target moved more quickly or the length of the digit sequence increased, performance gradually became poorer, and when the target moved more slowly or the digit sequence was shorter, performance improved, but as is clear from Figure 9.7, the effect of changing demand in a single task was exactly the

same for the patient group as it was for the other two groups. In a second experiment, the three groups were asked to perform the two tasks both set at a very easy level according to their own ability. Here, the patients had a problem in performing two easy tasks, while neither of the healthy groups showed a difference between single- and dual-task performance. Therefore, the patients showed a problem in carrying out two tasks at the same time, even when the tasks were extremely easy for them, and the effect of increasing demand had exactly the same effect on the patients as it did on the healthy participants. So it is the ability to perform two tasks at the same time that is a problem for the patients, and it is not an effect of overall demand on their damaged cognitive system.

Figure 9.7



Tracking accuracy as speed of target is gradually reduced or increased relative to the measured tracking speed (span) for each individual in a group of Alzheimer patients, age-matched healthy controls, and younger healthy controls. (Data reproduced from Logie et al., 2004.)

The specific impairment of dual-task ability in Alzheimer patients has now been shown in a range of other studies (e.g., Baddeley, Baddeley, Bucks, & Wilcock, 2001; Della Sala, Cocchini, Logie, & MacPherson, 2010; MacPherson, Della Sala, Logie, & Wilcock, 2007; Ramsden, Kinsella, Ong, & Storey, 2008; Sebastian, Menor, & Elosua, 2006). Indeed, Baddeley et al. (2001) contrasted tests of focal and sustained attention, inhibition, and divided attention, or dual task. Focal attention, measured with simple and choice response time tests was shown to be broadly intact in patients with Alzheimer's disease, in contrast to the clear impairment of divided attention when dual tasking.

Inhibition and Dementia

The patients in the Baddeley et al. (2001) study also appeared to have difficulty inhibiting irrelevant visual stimuli. In their comprehensive review of cognitive deficits in Alzheimer's disease, Perry and Hodges (1999) identified inhibition as being a particular difficulty for such patients. In a later review, Amieva, Phillips, Della Sala, and Henry (2004) noted that some studies had shown dementia-related impairments in tasks requiring control of inhibition, such as the Stroop task in which participants are asked to inhibit their natural tendency to read a color word (e.g., RED) but instead to say aloud the color of the ink in which the word is printed (e.g., for the word "RED" printed in green ink, the correct response is "green"). However, they also noted a range of studies showing little or no impairment in tasks involving automatic inhibition of responses such as the tendency not to look back at a location that has been fixated a very short time before, known as inhibition of return. In summary, it appears that focal attention and automatic inhibition are intact in Alzheimer's disease, but controlled inhibition and divided attention are not. Amieva et al. (2004) concluded that some forms of attention can be damaged while others remain intact, indicating in turn that there are different forms of attention in the healthy brain that can be differentially impaired by brain damage. This conclusion is further reinforced by the observation from earlier sections of this chapter that spatial attention is specifically impaired in neglect patients, in contrast with the impairment of divided attention and controlled inhibition in Alzheimer patients. Attention is not a single entity or concept.

A recent set of studies has examined the phenomenon of “closing-in behavior” in dementia patients (e.g., Ambron, McIntosh, Allaria, & Della Sala, 2009; Lee, Chin, Kang, Kim, Park, & Na, 2004). Here, Alzheimer patients are asked to copy a simple drawing on a sheet of paper, but they have a tendency to produce their drawing very close to, or even on top of, the item that they are copying. This has been interpreted either as their attention being attracted toward the existing drawing on the page (or a failure to inhibit this tendency) or as a limitation of their visuospatial working memory that might be needed to keep parts of the target item in mind while they look at and produce their own drawing or to switch their gaze between their drawing and their target. Although the phenomenon has been known in clinical settings for a long time (Mayer Gross, 1935), there are relatively few systematic empirical studies and theoretical accounts for the phenomenon at an early stage. However, as Ambron et al. (2009) note, it does appear to be linked with a combination of failures of attention and of visuospatial working memory.

Disorders of Sustained Attention

A further form of attention that has been studied in Alzheimer patients is vigilance or sustained attention. Here, the participant is asked to carry out a repetitive task over an extended period and has to detect the presence of an infrequently occurring event. The capacity to maintain attention over long intervals in order to detect infrequent signals assumed considerable practical importance during military conflicts in the mid 20th century, with the development of sonar and radar as submarine and aircraft detection devices. Performance in the radar/sonar operators tended to show a steady decrement over time, a finding that led to a considerable amount of research and contrasting interpretations as to its cause (see Broadbent, 1958, for a review). Parasuraman (1979, 1984) suggested that the decrement in sustained attention occurred in those conditions where participants had to hold information in memory for comparison with the signal that they were trying to detect, but it typically did not occur when participants had to make a simple perceptual judgement without any memory load. Parasuraman (1979) asked healthy participants to discriminate successive or simultaneous tones of different volume or light flashes of different intensity presented over periods of 45 minutes. Only the successive discrimination tasks resulted in progressive deterioration of performance. This led Parasuraman to argue that when vigilance tasks make ongoing demands on memory, this constitutes additional cognitive effort. A successive discrimination task involves retaining information in memory and so involves cognitive effort. Therefore, performance may begin to degrade earlier for a vigilance task with an ongoing memory load than it would in a task where the target and the comparison stimulus are presented together (simultaneous discrimination).

Sustained Attention in Dementia

Sustained attention deficits have been reported in clinical assessments of patients with Alzheimer’s disease (e.g., Della Sala & Logie, 1993; Parasuraman & Haxby, 1993; Spinnler, 1991). Alexander (1973) reported that dementia patients showed poorer overall detection of letter targets in a string of letters than did age-matched controls, but neither group showed a further drop in detection performance over time. A similar lack of difference in sustained attention performance between dementia patients and age-matched controls was reported by Nebes and Brady (1993) and by Brazzelli, Cocchini, Della Sala, and Spinnler (1994). Parasuraman and Haxby (1993) concluded from the results available at that time that Alzheimer patients show no decrement in vigilance if the task involves perceptual discrimination, and the differential impairments in the patients is found only when there is some additional task demand.

As discussed in the section on divided attention, there are problems with interpreting an observed impairment as resulting from high task demand or difficulty. This interpretation immediately leads to the question of what is the source of the difficulty. It could be that there is too much information, or this information is too complex, or it is being presented at too fast a rate for the capacity of the cognitive system,

whether that cognitive system is damaged or healthy. However, in the case of vigilance tasks, the whole point is that the task and the information are very simple and the information to which the participant has to respond is presented at a rather slow rate. Another possible source of difficulty could be that the information is degraded because of visual interference with the signal or noise in an auditory channel, or because the participant has poor vision or hearing. However, none of these problems were present in the experiments studying vigilance in dementia patients.

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Baddeley, Cocchini, Della Sala, Logie, and Spinnler (1999) noted that there is no a priori reason why the Parasuraman (1979) perceptual discrimination between two tones or flashes presented at the same time should involve a lower task demand than a discrimination in which a second item presented on its own is compared to a previously presented item that is held in memory. If the second condition leads to a vigilance decrement while the first does not, then this indicates a specific additional cognitive cost of storing information in short-term memory.

Baddeley et al. (1999) followed an analogous procedure to that used in the dual-task experiments (Baddeley et al., 1986; Logie et al., 2004) to equate initial performance levels across conditions and between groups of dementia patients and of older controls. They used a discrimination task in which two lines were shown on the screen and participants were asked to decide whether the lines were the same length. The discrimination threshold was measured for each participant—that is, the smallest difference in line lengths at which each participant could detect that the lengths were different. This ensured that the discrimination task itself was set at the same level of difficulty for all of the participants. In the perceptual task, both lines were presented simultaneously. In the memory condition, the second line was presented on its own and its length was to be compared with the first line of the pair held in memory. Results showed that the healthy older group showed a progressive drop over time, reaching a peak of around 10% reduction in performance after 30 minutes of the memory-based task, but showed no drop over that same period in the perceptual task. The Alzheimer patients also showed a progressive drop in the memory condition but reached a 24% reduction over just 15 minutes of performing the task. They also showed a drop of around 7% in the perceptual task over the same period. Given that the initial levels of performance were equated by adjusting task demands for each individual, this suggests that there is a specific impairment in sustained attention in Alzheimer patients, and this impairment is particularly evident when they have to carry a memory load while performing the vigilance task.

Sustained Attention and Frontal Lobe Damage

Damage to the prefrontal cortical areas of the brain are often associated with disorders of what are referred to as executive functions (e.g., Rabbitt, 1997), including inhibition, task switching, and focusing attention. The discussion so far in this chapter has considered possible links between damage to the frontal lobes with the progression of Alzheimer's disease, and impairments in divided and sustained attention. Patients with focal damage to the frontal lobes as a result of a stroke, or from neurosurgery, or from a serious head injury also appear to show impairments of attention. The most common clinical reports are that they exhibit "perseveration" in which there is a tendency to repeat the same action, confabulation in which they tend to generate reports of events that could not possibly have occurred, and a lack of inhibition such as "utilization behavior" in which they find it impossible to avoid picking up or handling objects placed in front of them. They might also have a tendency to say things that are insulting or rude because they cannot prevent themselves from doing so. Such patients have been shown to be unable to perform a range of planning tasks, and have a particular problem in multitasking, for example, planning and carrying out a set of errands in a shopping mall (Shallice & Burgess, 1991).

Although the impairments mentioned earlier are observed following frontal lobe damage, no one patient tends to exhibit all of these difficulties, and this has led to the suggestion mentioned earlier when

discussing the impact of Alzheimer's disease, that there are different kinds of attention that can be differentially impaired depending on the location and extent of the brain damage. For example, Simons, Marieke, Schölvink, Gilbert, Frith, and Burgess (2006) demonstrated that an area at the very front of the frontal lobes, Brodmann area 10, appears to be involved in biasing attention to external events or to internal, mental events. They have suggested that this area plays a crucial role in prospective memory tasks which require participants to be aware of external or exogenous cues, or to mental or endogenous cues that act as reminders to perform particular actions. If the bias is to exogenous cues but the relevant current cue is endogenous, then this will result in a prospective memory failure.

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One attentional impairment that does appear to be common to patients with frontal lobe damage is what has been referred to as "goal neglect" (e.g., Duncan, Elmslie, Williams, Johnson, & Freer, 1996). Here, the patient appears to ignore an instruction or particular aspect of a task when he or she has to perform it, even if the patient knows what the action is. So, for example, a patient might be asked to hold up his or her hand when a light comes on, and fail to do so. Yet he or she can say that when the light comes on he or she should hold up his or her hand. This neglect of current goals could be one reason why frontal lobe patients are poor at multitasking; they might know what the tasks are that they have to perform, but they neglect to perform them. Duncan and colleagues suggest that the action that is neglected loses out in a competition between a variety of mental activities or physical actions that the patient is asked to perform. As the set of goals or tasks becomes more complex, some of these are given priority reflected in a bias toward performing the higher priority tasks at the expense of others. Duncan has extended this idea to suggest that goal neglect might be characteristic of healthy individuals and linked to general mental capacity or general intelligence "g." Attending to multiple tasks or swapping attention between tasks is then more efficient for people who have greater mental capacity, who in turn will be less prone to show goal neglect. Given that prefrontal areas are thought to be important in supporting the planning of goals and switching attention between tasks, as well as switching attention between the external world and the mental world (Simons et al., 2006), then damage to these areas will be associated with an inability to deal with increasing task complexity, and hence goal neglect (Duncan et al., 2008).

Conclusions

This discussion of disorders of attention should make clear that it would be more appropriate to refer to disorders of different kinds of attention. Impairments of spatial attention in unilateral spatial perceptual neglect appear to be rather different from impairments associated with representational neglect, and both differ from the impairments of divided attention or of inhibition in Alzheimer's disease. These, in turn, only partially overlap with the lack of inhibition and goal neglect in patients suffering from frontal lobe damage. It is equally clear that there are striking differences between failures of attention to the outside world and failures of attention to the mental world in addition to a failure to switch attention between them and the operation of online cognition within the mental workspace (Logie, 2011; Logie & van der Meulen, 2009). There are other disorders of attention linked with, for example, healthy ageing (e.g., Naveh-Benjamin, Moscovitch, & Roediger, 2001), or syndromes such as attention-deficit disorders in children (e.g., Brown, 2005), or everyday lapses of attention leading to human errors in healthy adults (e.g., Reason, 1990), and discussion of each of these topics would require several chapters of this book.

Given the plethora of attentional disorders, is there some coherent theory or framework that might encompass all or most of these disorders, or are we really forced to come up with separate explanations for each of them? The human brain is certainly complex and it can break down in complex ways. It also has sufficient plasticity to compensate for some forms of damage, and when we observe a particular pattern of impairment in a brain-damaged individual, it is not always clear whether we are observing the result of the damage or the result of using some alternative cognitive system that relies on remaining intact brain

networks to perform the same task. Duncan's ideas of goal neglect coupled with biased competition between brain networks offers one possible general framework for attentional disorders. These ideas are also consistent with the long-standing view of the brain activity comprising differential levels of activation of different neuroanatomical networks, with the highest level of activation being given the highest priority (Selfridge, 1959; Selfridge & Neisser, 1960), the difference now being that this differential activity can be demonstrated rather than assumed (see review in Duncan, 2006). However, this cannot be the whole story given that the brain can process information in parallel, while focused and/or sustained attention might be elsewhere, and inhibitory processes can operate largely automatically. What has been discovered thus far appears to suggest that understanding both intact forms of attention and how they break down will require a combination of theories of controlled and automatic attention and theories of temporary storage and manipulation of material in the mental workspace of working memory.

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